

Research Experience

- **MADS&P — Security and Privacy Research Group** Madison, WI
Graduate Research Assistant Dec 2022 - present
 - My research focuses on differential privacy and learning theory, with an emphasis on problems that are relevant to real-world systems for prediction and resource allocation.
 - Advisor: Professor Kassem Fawaz
- **MetroSets** Tucson, AZ
Undergraduate Research Assistant Sep 2019 - Jan 2021
 - Designed and implemented a novel system for set visualization using the metro map metaphor, with sets drawn as subway lines and elements drawn as stations. Subsequently conducted a controlled user study which found that MetroSets outperformed other publicly available tools.
- **Authorship Attribution** Tucson, AZ
Honors Thesis May 2020 - Jul 2021
 - Studied the robustness of machine learning systems designed to classify the author of a binary file. Designed and implemented an adversarial attack on state-of-the-art tools by using non-standard compiler optimizations to obscure an author’s stylistic fingerprint.

Conference Publications

Jacobsen, B., Kohli, N. (2026). *Privacy, Prediction, and Allocation*. 7th annual Symposium on Foundations of Responsible Computing (FORC).

Jacobsen, B., Fawaz, K. (2026). *Prediction with Expert Advice under Local Differential Privacy*. Fourteenth International Conference on Learning Representations (ICLR). (Earlier version presented at the Theory and Practice of Differential Privacy Workshop Series 2025 (TPDP)).

Jacobsen, B., Fawaz, K. (2025). *Private Continual Counting of Unbounded Streams*. Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS).

Journal Publications

Waller, M., **Jacobsen, B.**, Kobourov, S., and Nollenburg, M. (2021). *On the Readability of Abstract Set Visualizations*. IEEE Transactions on Visualization and Computer Graphics, 27(2). (Selected for TVCG special issue as one of the top 5 papers from IEEE PacificVis 2021.).

Jacobsen, B., Waller, M., Kobourov, S., and Nollenburg, M. (2020). *MetroSets: Visualizing Sets as Metro Maps*. IEEE Transactions on Visualization and Computer Graphics, 27(6). (Presented at IEEE VIS 2020.)

Workshop Publications

Jacobsen, B., Rahaman, S., and Debray, S. (2021) *Optimization to the Rescue: Evading Binary Code Stylometry with Adversarial Use of Code Optimizations*. Proceedings of the 2021 Research on offensive and defensive techniques in the Context of Man At The End (MATE) Attacks.

Education

- University of Wisconsin - Madison** Madison, WI
 - *PhD Student, Computer Science* Expected 2027
 - *M.S., Computer Science* May 2024
- University of Arizona (Honors College)** Tucson, AZ
 - *B.A., Mathematics, minor in Computer Science, Summa Cum Laude* May 2021
- Central New Mexico Community College** Albuquerque, NM
 - *A.S. with Highest Honors, Mathematical Sciences* August 2019

Teaching Experience

- Graduate Lecturer**
 - *University of Wisconsin - Madison* Jan 2026 - May 2026
 - Sole instructor for CS642 (Introduction to Information Security)
 - Designed course materials, activities and assessments, introducing advanced undergraduate students to major topics in systems security, cryptography, and privacy.
- Part-Time Faculty in Computer Science**
 - *Madison Area Technical College* Aug 2025 - Dec 2025
 - Part of the first cohort of Madison College Teaching Fellows, a professional development program for graduate students interested in teaching careers.
 - Collaborated with faculty mentor to incorporate active learning in an introductory Java class.
 - Designed, presented and facilitated 4 hours of in-person lecture and labs each week.
- Head Teaching Assistant**
 - *University of Wisconsin - Madison* Aug 2025 - Dec 2025
 - Designed autograding infrastructure, oversaw assignment development and managed fellow TAs for CS300 (Programming II), a ~500 student required course.
- Graduate Teaching Assistant**
 - *University of Wisconsin - Madison* Aug 2021 - May 2025
 - Held office hours, graded assessments and assisted students across a wide variety of courses:
 - * CS240 (Introduction to Discrete Mathematics, for first-year CS majors)
 - * CS300 (Programming II, for first-year CS majors)
 - * CS320 (Data Science Programming II, for non-CS majors interested in programming)
 - * CS642 (Introduction to Information Security, senior-level elective; 3 semesters)
 - * CS760 (Machine Learning, for graduate students; 2 semesters)

Additional Work Experience

- **Research Data Support Specialist**
University of Arizona *June 2021 - August 2021*
- **Undergraduate Research Assistant**
University of Arizona *Sep 2019 - May 2021*
- **Programming Tutor**
Private *June 2020 - August 2021*
- **Software Consultant**
Fyberloom Inc. *June 2021 - July 2021*

Awards and Honors

NeurIPS Travel Grant	2025
Madison College Teaching Fellowship	2025
UA College of Science Excellence in Undergraduate Research Award (Overall Winner)	2021
CRA Outstanding Undergraduate Researcher Honorable Mention	2020
Phi Beta Kappa Honors Society	2021-Present
Phi Theta Kappa Honors Society	2018-Present
Phi Theta Kappa Transfer Scholarship (\$24,000)	2019-2020
Galileo Circle Scholarship (\$1,000)	2020

Invited Talks

- **Artificial Intelligence in Research: Ethical Challenges and Considerations**
Panelist in presentation for graduate students and faculty in nuclear engineering *March 2025*
- **Contextual Integrity for Computer Scientists**
Presentation for Wisconsin Privacy and Security Group *December 2024*
- **Legal Risks in Security Research**
Presentation for Madison Security & Privacy Seminar *October 2024*
- **Introduction to Security & Privacy in Machine Learning**
Guest Lecture for CS642 — Introduction to Information Security *April 2023*

Skills

Programming Languages: Python, Java, MATLAB

Natural Languages: English, Mandarin (intermediate)

Operating Systems: Linux (Arch, Ubuntu), Windows

Tools: $\text{\LaTeX}$, Git, Docker, VirtualBox, NumPy, Pandas, Matplotlib, PyTorch, TensorFlow